

Reg No.:

Name:

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) / S4 (PT) (R,S) / S2 (PT) (R,S) Examination June 2023 (2019 Scheme)

Course Code: MAT 204

Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS

Duration: 3 Hours

Max. Marks: 100

(Statistical Tables are allowed)

PART A

(Answer all questions; each question carries 3 marks)

Marks

The following table gives the probabilities that a certain computer will malfunction 0,1,2,3,4,5 or 6 times on any day.

No. of Malfunctions (x)	0	1	2	3	4	5	6
Probability f(x)	0.17	0.29	0.27	0.16	0.07	0.03	0.01

Find the mean and variance of the probability distribution.

3

Let X and Y have the following probability distribution.

X\Y	1	2
1	0.15	0.10
2	0.30	0.20
3	0.15	0.10

Check whether X and Y are independent or not.

3

Derive the mean and variance of an Exponential distribution.

3

The time for a superglue to set can be treated as a random variable having a normal distribution with mean 30 seconds. Find its standard deviation if the probability is 0.20 that it will take on a value greater than 39.2 seconds.

3

State any three properties of autocorrelation function.

3

Find the power spectral density of the WSS process with autocorrelation function

$$R(\tau) = 3e^{-2|\tau|}$$

Using Newton Raphson Method find a positive root of $x^4 - x - 10 = 0$ correct to 3 decimal places.

3

Using Trapezoidal rule evaluate the integral $\int_1^2 \frac{dx}{1+x^2}$ with $h = 0.5$.

9/

By method of least squares, Fit the straight line of the form $y = ax + b$ for

x	1	2	3	4	5
y	5	7	9	10	11

10/

Using Second order Runge-Kutta Method, evaluate $y(0.1)$ with $h=0.1$ from $\frac{dy}{dx} =$

$$y - \frac{2x}{y}, y(0) = 1.$$

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11/ a) The probability mass function of a discrete random variable is $f(x) = kx$, $x = 1, 2, 3$ where k is a positive constant. Find (i) the value of k (ii) $P(X \leq 2)$ (iii) $E(X)$ and (iv) $\text{Var}(1-X)$. 7
- b) If X follows a Poisson distribution such that $P(X = 1) = 3/10$ and $P(X = 2) = 1/5$. Find $P(X = 4)$ and $P(X = 0)$. 7
- 12/ a) The probability of a bomb hitting a target is $\frac{1}{5}$. Two bombs are enough to destroy a bridge. If 6 bombs are aimed at the bridge, find the probability that the bridge is destroyed. 7
- b) If the joint probability density function of random variables X and Y is given by $f(x, y) = c(2x + 3y)$, $x = 0, 1, 2$; $y = 1, 2, 3$. Find (i) the value of c (ii) the marginal probability distributions of X and Y (iii) the probability distribution of $X+Y$. 7

Module -2

- 13/ a) The cumulative distribution function of a continuous variable X is given by 7
- $$F(x) = \begin{cases} 0, & x \leq 2 \\ c(x-2), & 2 < x < 6 \\ 1, & x \geq 6 \end{cases}$$
- Find (i) p.d.f (ii) value of c (iii) $P(1 \leq X \leq 5)$
- (b) Buses arrive at a specified stop at 15 min. intervals starting at 7 a.m. If a passenger arrives at the stop at a random time that is uniformly distributed between 7:00 and 7:30 a.m., find the probability that he waits (i) less than 5 minutes for a bus (ii) at least 12 minutes for a bus. 7
- 14/ a) Examine whether the random variables X and Y are independent, whose joint probability density function is $f(x, y) = xe^{-x(y+1)}$, $0 < x$; $y < \infty$ 7

- b) A distribution with unknown mean μ has variance equal to 1.5. By using central limit theorem, find how large a sample should be taken in order that the probability will be at least 0.95 that the sample mean will be within 0.5 of the population mean. 7

Module -3

- 15 a) $X(t)$ is a random process defined by $X(t) = A \cos(\omega t + \theta)$ where ω is a constant and A and θ are independent random variables uniformly distributed in $(-k, k)$ and $(-\pi, \pi)$ respectively. (i) Find the mean, (ii) autocorrelation (iii) Is $X(t)$ WSS? 7

- (b) If $X(t)$ is a WSS process with autocorrelation $R_X(\tau) = \frac{\tau^2 + 44}{\tau^2 + 4}$. Find the mean and variance. 7

- 16 a) Assume that $X(t)$ is a random process defined as follows: $X(t) = A \cos(2\pi t + \phi)$ where A is a zero-mean normal random variable with variance $\sigma_A^2 = 2$ and ϕ is uniformly distributed random variable over the interval $-\pi \leq \phi \leq \pi$. A and ϕ are statistically independent. Let the random variable Y be defined as $Y = \int_0^1 X(t) dt$. Determine (i) the mean of Y (ii) the variance of Y . 7

- b) If people arrive at a book stall in accordance with a Poisson process with a mean rate of 3 per minute, find the probability that the interval between 2 consecutive arrivals is (i) more than 1 minute, (ii) between 1 minute and 2 minutes (iii) 4 minutes or less. 7

Module -4

- 17 a) Solve the equation $3x + \sin x - e^x = 0$ by Regula Falsi Method. 7
- b) Using Lagrange's interpolation method find $y(4)$, Given the following data. 7

x	1	3	5	7
y	24	120	336	720

- (18) a) Evaluate $y(2)$ from the following data using Newton's forward difference method. 7

x	0	5	10	15
y	14	379	1444	3584

- b) Find $y(3)$ Using Newton's divided difference formula, Given $y(1) = -26, y(2) = 12, y(4) = 256, y(6) = 844$. 7

Module -5

- 19 a) Solve the equations $4x + 2y + z = 14, x + 5y - z = 10, x + y + 8z = 20$, using Gauss-Siedel Method. 7

- b) Using Euler's method, solve $\frac{dy}{dx} = x + y + xy$, given $y(0) = 1$, evaluate y at $x = 0.1$ by taking $h = 0.05$. 7
- 20 a) Apply fourth order Runge Kutta Method to find $y(0.1)$ for the initial value problem $\frac{dy}{dx} = x + y^2, y(0) = 1$. 7
- b) Solve $\frac{dy}{dx} = x^2(1 + y)$ for $x = 1.4$ using Adam's Moulton Method, given $y(1) = 1, y(1.1) = 1.233, y(1.2) = 1.548, y(1.3) = 1.979$. 7

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B.Tech Degree $S_4(R, S) / S_4(PT)(R, S) / S_2(PT)(R, S)$
Examination June 2023 (2019 scheme)

Course Code: MAT 204

course name: Probability, Random Processes and Numerical Methods

1. The following table gives the probabilities that a certain computer will mal function 0, 1, 2, 3, 4, 5 or 6 times on any day.

No of Malfunctions (x)	0	1	2	3	4	5	6
Probability $f(x)$	0.17	0.29	0.27	0.16	0.07	0.03	0.01

Find the mean and variance of probability distribution

Ans) Mean = $\sum x f(x) = 0 \times 0.17 + 1 \times 0.29 + 2 \times 0.27 + 3 \times 0.16$
 $+ 4 \times 0.07 + 5 \times 0.03 + 6 \times 0.01$
 $= 0 + 0.29 + 0.54 + 0.48 + 0.28 + 0.16$
 $+ 0.06$
 $= \underline{\underline{1.8}}$

$$E(x^2) = \sum x^2 f(x)$$

$$= 0^2 \times 0.17 + 1^2 \times 0.29 + 2^2 \times 0.27 + 3^2 \times 0.16 + 4^2 \times 0.07 + 5^2 \times 0.03 + 6^2 \times 0.01$$

$$= 0.29 + 1.08 + 1.44 + 1.12 + 0.75 + 0.36$$

$$= \underline{\underline{5.04}}$$

$$V(x) = E(x^2) - [E(x)]^2$$

$$= 5.04 - (1.8)^2 = 5.04 - 3.24$$

$$= \underline{\underline{1.8}}$$

2. Let x and y have the following probability distribution

$x \backslash y$	1	2.
1	0.15	0.10
2	0.30	0.20
3	0.15	0.10.

check whether x and y are independent or not.

$x \backslash y$	1	2	P_x
1	0.15	0.10	0.25
2	0.30	0.20	0.50
3	0.15	0.10	0.25
P_y	0.60	0.40	1

g) x and y are independent then

$$P_x(x) P_y(y) = P(x, y).$$

$$0.60 \times 0.25 = 0.15$$

$$0.40 \times 0.25 = 0.1$$

$$0.60 \times 0.50 = 0.3$$

$$0.40 \times 0.50 = 0.2$$

$$0.60 \times 0.25 = 0.15$$

$$0.40 \times 0.25 = 0.1$$

$\therefore x$ and y are independent

3 Derive the mean and variance of Exponential distribution.

Ans) Mean = $E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot e^{-x} dx$
 $= 1 \int_0^{\infty} x e^{-x} dx.$

$$1 \left[\frac{x e^{-1x}}{-1} - 1 \cdot \frac{e^{-1x}}{-1^2} \right]_0^{\infty}$$

$$= \underline{\underline{1/1}}$$

$u = x$	$v = e^{-1x}$
$u' = 1$	$v' = \frac{e^{-1x}}{-1}$
$u'' = 0$	$v'' = \frac{e^{-1x}}{1^2}$

$$\int uv = uv_1 - u'v_2 + \dots$$

To find Variance

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \frac{e^{-1x}}{1} dx$$

$$= 1 \int_0^{\infty} x^2 e^{-1x} dx$$

$$= 1 \left[\frac{x^2 e^{-1x}}{-1} - 2x \frac{e^{-1x}}{1^2} + \frac{2 e^{-1x}}{-1^3} \right]_0^{\infty}$$

$$= 1 \left[(0 - 0 + 0) - (0 - 0 + \frac{2}{-1^3}) \right]$$

$$= 1 \left(\frac{2}{1^3} \right) = \frac{2}{1^2}$$

$$V(x) = \frac{2}{1^2} - \left(\frac{1}{1} \right)^2 = \underline{\underline{\frac{1}{1^2}}}$$

4. The time for a superglue to set can be treated as a random variable having a normal distribution with mean 80 seconds. Find its standard deviation if the probability is 0.20 that it will take on a value greater than 87.2 seconds.

$$p(x > 39.2) = 0.20$$

$$p\left(\frac{x - \mu}{\sigma} > \frac{39.2 - \mu}{\sigma}\right) = 0.20$$

$$p\left(z > \frac{39.2 - 30}{\sigma}\right) = 0.20$$

$$p(z > \frac{9.2}{\sigma}) = 0.20$$

$$1 - p(z \leq \frac{9.2}{\sigma}) = 0.20$$

$$1 - 0.20 = p(z \leq \frac{9.2}{\sigma}) = F\left(\frac{9.2}{\sigma}\right)$$

$$0.80 = F\left(\frac{9.2}{\sigma}\right)$$

$$\frac{9.2}{\sigma} = 0.85$$

$$\sigma = \frac{9.2}{0.85}$$

$$\sigma = 10.8$$

2. State any 3 properties of auto correlation function.

1. The mean square value of a random process can be obtained from the autocorrelation function.
2. Auto correlation fn is even.
3. Maximum value of autocorrelation $R(z)$ is attained at z_0 .

Find the power spectral density of the WSS process with auto correlation function
 $R(\tau) = 3e^{-2|\tau|}$

$$S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-i\omega\tau} d\tau.$$

$$= \int_0^{\infty} 3e^{-2\tau} e^{-i\omega\tau} d\tau.$$

$$= 3 \int_0^{\infty} e^{-(2+i\omega)\tau} d\tau = 3 \int_0^{\infty} e^{-\tau[2+i\omega]} d\tau$$

$$= 3 \left[\frac{e^{-\tau[2+i\omega]}}{-(2+i\omega)} \right]_0^{\infty} =$$

7. Using Newton Raphson Method find a positive root of $x^4 - x - 10 = 0$ correct to 3 decimal places.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$\therefore$ Root lies b/w 1 and 2.

$$x_0 = \frac{a+b}{2} = \frac{1+2}{2} = \frac{3}{2} = 1.5$$

$$\begin{aligned} f(x) &= x^4 - x - 10. \\ f(1) &= 1 - 1 - 10 = -10 \\ f(2) &= 2^4 - 2 - 10 \\ &= 16 - 2 - 10 \\ &= 4. \end{aligned}$$

$\therefore f(1)$ and $f(2)$ are of opposite signs

put $n=0$

$$\begin{aligned}x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\&= 1.5 - \frac{f(1.5)}{f'(1.5)}\end{aligned}$$

$$= 1.5 - \frac{-6.4375}{12.5}$$

$$= 1.5 + 0.515$$

$$= \underline{\underline{2.015}}$$

$$f(x) = x^4 - x - 10$$

$$f'(x) = 4x^3 - 1$$

put $n=1$, $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$= 2.015 - \frac{f(2.015)}{f'(2.015)}$$

$$= 2.015 - \frac{4.4704270506}{31.7254135}$$

$$= \underline{\underline{1.874}}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 1.874 - \frac{f(1.874)}{f'(1.874)}$$

$$= \underline{\underline{1.856}}$$

$$\begin{aligned}
 x_4 &= x_3 - \frac{f(x_3)}{f'(x_3)} \\
 &= 1.856 - \frac{f(1.856)}{f'(1.856)} \\
 &= \underline{\underline{1.856}}
 \end{aligned}$$

8. Using Trapezoidal rule evaluate the integral

$$\int_1^2 \frac{dx}{1+x^2} \text{ with } h=0.5$$

1	1.5	2
f_0	f_1	f_2

$$\int_1^2 \frac{dx}{1+x^2} = \frac{h}{2} [f_0 + f_2 + 2f_1]$$

$$= \frac{0.5}{2} [0.5 + 0.2 + 2 \times 0.3077]$$

$$= \underline{\underline{0.3289}}$$

$$f(x) = \frac{1}{1+x^2}$$

$$f_0 = \frac{1}{1+1^2} = 0.5$$

$$f_1 = \frac{1}{1+(1.5)^2} =$$

$$\dots 0.3077$$

$$f_2 = \frac{1}{1+2^2} = 0.2$$

9 By the method of least squares, Fit the straight line of the form $y = ax + b$ for

x	1	2	3	4	5
y	5	7	9	10	11

st. line eqn is $y = ax + b$. — (A)

$$\sum y = a \sum x + nb. \quad \text{--- (1)}$$

$$\sum xy = a \sum x^2 + b \sum x \quad \text{--- (2)}$$

$$(1) \Rightarrow 42 = a \cdot 15 + 5b.$$

$$(2) \Rightarrow 141 = 55a + 15b$$

$$(1) \times 3 \Rightarrow 126 = 45a + 15b. \quad \text{--- (3)}$$

$$141 = 55a + 15b. \quad \text{--- (2)}$$

$$(3) - (2) \quad 126 - 141 = -10a.$$

$$-15 = -10a.$$

$$a = \frac{15}{10} = \underline{\underline{1.5}}$$

$$(1) \Rightarrow 42 = 15 \times 1.5 + 5b.$$

$$5b = 42 - 15 \times 1.5.$$

$$b = 3.9.$$

$\therefore$ Required fit $y = \underline{\underline{1.5x + 3.9}}$

10. Using second order Runge-Kutta method, evaluate $y(0.1)$ with $h = 0.1$ from $\frac{dy}{dx} = y - \frac{2x}{y}$, $y(0) = 1$.

Ans) $f(x, y) = y - \frac{2x}{y}$

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f(x_n + h, y_n + k_1)$$

$x_0 = 0$	0.1
$y_0 = 1$	$y_1 ?$

$$x_0 = 0$$

$$y_0 = 1$$

$$f(x, y) = y - \frac{x}{y}$$

$$y_1 = y_0 + \frac{1}{2} (k_1 + k_2), \quad k_1 = h f(x_0, y_0)$$

$$= 0.1 f(0, 1)$$

$$= 0.1 \times 1 = 0.1$$

$$k_2 = h f(x_0 + h, y_0 + k_1)$$

$$= 0.1 f(0 + 0.1, 1 + 0.1)$$

$$= 0.1 f(0.1, 1.1)$$

$$= 0.1 f(0.1, 1.1)$$

$$= 0.1 \left[0.1 - \frac{0.2 \times 0.1}{1.1} \right]$$

$$= 0.1 \left[0.1 - 0.0181818 \right]$$

$$= 0.0081818$$

$$= 1 + \frac{1}{2} [0.1 + 0.0081818]$$

$$= \underline{\underline{1.0459}}$$

Module 1

11) The probability mass function of a discrete r.v is $f(x) = kx$, $x = 1, 2, 3$ where k is a positive constant. Find ① the value of k ② $P(x \leq 2)$ ③ $E(x)$ and

④ $V(x)$

Ans)

x	1	2	3
$f(x)$	k	$2k$	$3k$

given f is p.m.f

$$\therefore \sum f(n) = 1$$

$$k + 2k + 3k = 1.$$

$$6k = 1$$

$$k = \frac{1}{6}$$

$$p(x \leq 2) = p(x=1) + p(x=2)$$

$$= \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \underline{\underline{\frac{1}{2}}}$$

$$E(x) = \sum x f(n) = 1 \times f(1) + 2 \times f(2) + 3 \times f(3)$$

$$= 1 \times \frac{1}{6} + 2 \times \frac{2}{6} + 3 \times \frac{3}{6}$$

$$= \frac{1}{6} + \frac{4}{6} + \frac{9}{6}$$

$$= \frac{14}{6} = \frac{7}{3} = \underline{\underline{2.333}}$$

$$V(1-x) = (-1)^2 V(x) = V(x). \rightarrow V(1) = 0.$$

$$V(x) = E(x^2) - [E(x)]^2$$

$$\begin{cases} V(ax+b) = a^2 V(x) \\ V(1-x); a=-1 \\ \downarrow \\ b=1 \\ = (-1)^2 V(x) \end{cases}$$

$$E(x^2) = \sum x^2 f(n) = 1 \times f(1) + 4 \times \frac{2}{6} + 9 \times \frac{3}{6}$$

$$= \frac{1}{6} + \frac{8}{6} + \frac{27}{6}$$

$$= \frac{36}{6} = 6.$$

$$V(x) = 6 - \left(\frac{7}{3}\right)^2 = 0.555 = \underline{\underline{\frac{5}{9}}}$$

$$\therefore V(1-x) = V(x) = \underline{\underline{\frac{5}{9}}}$$

11 b) Q6 x follows a poisson distribution such that
 $p(x=1) = 3/10$ and $p(x=2) = 1/5$ Find $p(x=4)$
 and $p(x=0)$.

Ans)
$$p(x=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$p(x=1) = \frac{3}{10} \Rightarrow \frac{e^{-\lambda} \lambda^1}{1!} = \frac{3}{10} \quad \text{--- (2)}$$

$$e^{-\lambda} \lambda = \frac{3}{10}$$

$$p(x=2) = \frac{1}{5} \Rightarrow \frac{e^{-\lambda} \lambda^2}{2!} = \frac{1}{5} \quad \text{--- (3)}$$

$$e^{-\lambda} \lambda^2 = \frac{2}{5}$$

divide (3) by (2)

$$\frac{e^{-\lambda} \lambda^2}{e^{-\lambda} \lambda} = \frac{\frac{2}{5} \times \frac{10}{3}}{\frac{3}{10}} = \frac{4}{3}$$

$$\lambda = \underline{\underline{\frac{4}{3}}}$$

1.2-3-4

$$p(x=4) = \frac{e^{-4/3} \left(\frac{4}{3}\right)^4}{4!} = \underline{\underline{0.0347}}$$

$$p(x=0) = \frac{e^{-4/3} \left(\frac{4}{3}\right)^0}{0!} = \underline{\underline{e^{-4/3}}}$$

12a) The probability of a bomb hitting a target is $\frac{1}{5}$. Two bombs are enough to destroy a bridge. If 6 bombs are aimed at the bridge, find the probability that the bridge is destroyed.

$$p = \frac{1}{5} \quad q = \frac{4}{5} \quad n = 6.$$

$$\frac{3 \times 5}{1.2} = 15$$

$$p(x=x) = {}^nC_x p^x q^{n-x}$$

$$= {}^6C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{6-x}$$

$$0.6$$

$$p(x=2) = {}^6C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^{6-2=4}$$

$$= {}^6C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4$$

$$= \underline{\underline{0.24576}}$$

12b) If the joint probability density fn. of random variables x and y is given by $f(x,y) = c(2x+3y)$

$x = 0, 1, 2$ $y = 1, 2, 3$. Find (1) the value of c .

(2) the marginal probability distributions of x and y .

(3) The probability distribution of $x + y$

$x \backslash y$	1	2	3
0	$3c$	$6c$	$9c$
1	$5c$	$8c$	$11c$
2	$7c$	$10c$	$13c$

$$f(x, y) = c(2x + 2y)$$

$$x = 0, 1, 2$$

$$y = 1, 2, 3$$

given $f(x, y)$ is p.m.f.

$$\sum_x \sum_y f(x, y) = 1$$

$$\Rightarrow 3c + 6c + 9c + 5c + 8c + 11c + 7c + 10c + 13c = 1$$

$$72c = 1$$

$$c = 1/72$$

$$f_x(x) = 18/72, 24/72, 30/72$$

$$f_y(y) = 15/72, 24/72, 33/72$$

$x+y$	1	2	3	4	5
$f(x+y)$	$3/72$	$11/72$	$24/72$	$21/72$	$13/72$

Module 2

13 a) The cumulative distribution function of a continuous r.v. X is given by

$$F(x) = \begin{cases} 0 & x \leq a \\ c(x-a) & a < x < b \\ 1 & x \geq b \end{cases}$$

Find (1) p.d.f (2) value of c (3) $P(1 \leq x \leq 5)$

Ans. 1) $f(x) = \begin{cases} c & 2 < x < 6 \\ 0 & \text{otherwise} \end{cases}$

2) $\int f(x) dx = 1 \Rightarrow \int_2^6 c dx = 1 \Rightarrow c \left[x \right]_2^6 = 1$

$c(6-2) = 1 \Rightarrow 4c = 1 \Rightarrow c = \underline{\underline{1/4}}$

3) $P(1 \leq x \leq 5) = \int_1^5 c dx = c(x) \Big|_1^5 = \frac{1}{4} (5-1) = \frac{4}{4} = 1$

13b) Buses arrive at a specified stop at 15 min. intervals starting at 7 a.m. If a passenger arrives at the stop at a random time that is uniformly distributed between 7.00 and 7.30 a.m., find the probability that he waits 1) less than 5 minutes for a bus. 2) at least 10 minutes for a bus.

Ans. Uniform distribution $\Rightarrow f(x) = \begin{cases} 1/30 & 0 < x < 30 \\ 0 & \text{otherwise} \end{cases}$

1) probability of he waits less than 5 minutes for a bus = $P(10 \leq x \leq 15) + P(25 \leq x \leq 30)$

$$\begin{aligned}
 &= \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx \\
 &= \frac{1}{30} (x)_{10}^{15} + \frac{1}{30} (x)_{25}^{30} \\
 &= \frac{1}{30} (15-10) + \frac{1}{30} (30-25) \\
 &= \frac{5}{30} + \frac{5}{30} = \frac{10}{30} = \underline{\underline{\frac{1}{3}}}
 \end{aligned}$$

2) $p(\text{he waits atleast 12 minutes for a bus})$

$$= p(0 \leq x \leq 3) + p(15 \leq x \leq 18)$$

$$= \int_0^3 \frac{1}{30} dx + \int_{15}^{18} \frac{1}{30} dx$$

$$= \frac{1}{30} (x)_0^3 + \frac{1}{30} (x)_{15}^{18}$$

$$= \frac{1}{30} \cdot 3 + \frac{1}{30} (18-15)$$

$$= \frac{3}{30} + \frac{3}{30} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \underline{\underline{\frac{1}{5}}}$$

14 a) Examine whether the random variables x and y are independent, whose joint probability density function is $f(x, y) = x e^{-x(y+1)}$ $0 < x, y < \infty$

$$f_x(x) = \int_y f(x, y) dy = \int_0^{\infty} x e^{-xy} e^{-x} dy$$

14.6) A distribution with unknown mean μ has variance equal to 1.5. By using central limit theorem, find how large a sample should be taken in order that the probability will be atleast 0.95 that the sample mean will be within 0.5 of the population mean.

soln) Given mean = μ
 $E(X_i) = \mu$

Variance = 1.5

$V(X_i) = 1.5$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad (\text{central limit theorem})$$

$$P(|\bar{X} - \mu| < 0.5) \geq 0.95$$

$$P\left(\left|\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right| < \frac{0.5}{\sigma/\sqrt{n}}\right) \geq 0.95$$

$$P(|Z| < \frac{0.5}{\sigma/\sqrt{n}}) \geq 0.95$$

$$P\left(0 < Z < \frac{0.5}{\sigma/\sqrt{n}}\right) \geq 0.95$$

$$\frac{0.5}{\sigma/\sqrt{n}} > 1.96 \Rightarrow$$

$$\frac{\sqrt{n} \cdot 0.5}{\sigma} > 1.96 \quad \text{2.}$$

$$\frac{n \cdot (0.5)^2}{\sigma^2} > (1.96)^2$$

$$\frac{n \times 0.25}{\text{variance}} > 3.8416$$

$$\frac{n \times 0.25}{1.5} > 3.8416$$

$$n > \frac{3.8416 \times 1.5}{0.25}$$

$$n > 23.0496$$

$\therefore$ least sample size is 24

Module 3.

- 15 a) $x(t)$ is a random process defined by $x(t) = A \cos(\omega t + \theta)$ where ω is a constant and A and θ are independent random variables uniformly distributed in $(-K, K)$ and $(-\pi, \pi)$ respectively. 1) Find the mean 2) auto correlation
3) Is $x(t)$ WSS? ?

Ans)

$$x(t) = A \cos(\omega t + \theta)$$

$$f(A) = \frac{1}{2K}, \quad -K \leq A \leq K.$$

$$g(\theta) = \frac{1}{2\pi}, \quad -\pi \leq \theta \leq \pi$$

$$E(A) = \frac{\alpha + \beta}{2} = \frac{-k + k}{2} = 0.$$

$$V(A) = \frac{(\beta - \alpha)^2}{12} = \frac{(k - -k)^2}{12} = \frac{k^2}{3}$$

$$E(A^2) = V(A) + (E(A))^2 = \frac{k^2}{3} + 0 = \frac{k^2}{3}$$

$$\begin{aligned} E(x(t)) &= E(A \cos(\omega t + \theta)) \\ &= E(A) \cos(\omega t + \theta) \\ &= 0. \end{aligned}$$

$$\begin{aligned} R_{xx}(t_1, t_2) &= E(x(t_1) x(t_2)) \\ &= E(A \cos(\omega t_1 + \theta) A \cos(\omega t_2 + \theta)) \\ &= E(A^2) (E(\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta))) \\ &= \frac{k^2}{3} E(\cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)) \end{aligned}$$

$$= \frac{k^2}{3} E \left[\frac{\cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta)}{2} \right]$$

$$= \frac{k^2}{3 \times 2} \int_{-\infty}^{\infty} [\cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta)] g(\theta) d\theta.$$

$$= \frac{k^2}{3} \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\theta) d\theta$$

$$= \frac{k^2}{b \cdot 2\pi} \left[\cos \omega(t_1 - t_2) \theta + \frac{\sin(\omega(t_1 + t_2) + 2\theta)}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{k^2}{12\pi} \left[\cos \omega(t_1 - t_2) \cdot \pi + \frac{\sin(\omega(t_1 + t_2) + 2\pi)}{2} - \cos \omega(t_1 - t_2)(-\pi) - \frac{\sin(\omega(t_1 + t_2) - 2\pi)}{2} \right]$$

$$= \frac{k^2}{12\pi} \left[2\pi \cos \omega(t_1 - t_2) + \frac{\sin \omega(t_1 + t_2)}{2} - \frac{\sin \omega(t_1 + t_2)}{2} \right]$$

$$= \frac{k^2}{12\pi} 2\pi \cos \omega(t_1 - t_2) = \underline{\underline{\frac{k^2}{b} \cos \omega(t_1 - t_2)}}$$

Since mean is constant, and auto correlation function depends only on time difference, then $x(t)$ is WSS.

15b. If $x(t)$ is a WSS process with autocorrelation $R_x(z) = \frac{z^2 + 4}{z^2 + 4}$. Find the mean & variance

$$R_X(z) = \frac{z^2 + 44}{z^2 + 4}$$

$$M_X^2 = \lim_{z \rightarrow \infty} R_X(z), \quad M_X = 1$$

$$E(X^2(t)) = R_X(0) = 44/4 = 11$$

$$\text{Variance} = 10.$$

169) Assume that $x(t)$ is a random process defined as follows: $x(t) = A \cos(2\pi t + \phi)$

where A is a ~~non~~ zero-mean normal random variable with variance $\sigma_A^2 = 2$

and ϕ is uniformly distributed r.v. over the interval $-\pi \leq \phi \leq \pi$. A and ϕ are

statistically independent. Let the random variable Y be defined as $Y = \int_0^1 x(t) dt$.

Determine 1) mean of Y 2) Variance of Y

$$X(t) = A \cos(2\pi t + \phi)$$

Ans)

$$\textcircled{1} E(X(t)) = E(A \cos(2\pi t + \phi))$$

$$E(X(t)) = E(A) E(\cos 2\pi t + \phi) = 0.$$

$$\sigma^2_{X(t)} = 1.$$

$$\begin{aligned} E(Y) &= E\left[\int_0^1 X(t) dt\right] \\ &= E\left[\int_0^1 A \cos(2\pi t + \phi) dt\right] \\ &= 0 \end{aligned}$$

$$\textcircled{2} \cdot \sigma^2_Y = \underline{\underline{E(Y^2) = 0.}}$$

16 b) If people arrive at a book stall in accordance with a Poisson process with a mean rate of 3 per minute, find the probability that the interval between 2 consecutive arrivals is $\textcircled{1}$ more than 1 minute $\textcircled{2}$ between 1 minute and 2 minutes $\textcircled{3}$ 4 minutes or less.

Ans)

Poisson process, $\lambda = 3.$

$$f(t) = 1e^{-1t} = 3e^{-3t}, \quad t > 0.$$

① probability of (more than 1 minute)

$$\begin{aligned} = p(t > 1) &= \int_1^{\infty} 3e^{-3t} dt = 3 \left[\frac{e^{-3t}}{-3} \right]_1^{\infty} \\ &= -[e^{-\infty} - e^{-3}] = -[0 - e^{-3}] \\ &= e^{-3} \end{aligned}$$

② probability of (between 1 minute and 2 minutes)

$$\begin{aligned} = p(1 < t < 2) &= \int_1^2 3e^{-3t} dt = 3 \left[\frac{e^{-3t}}{-3} \right]_1^2 \\ &= -[e^{-3t}]_1^2 = -[e^{-6} - e^{-3}] \\ &= \underline{\underline{e^{-3} - e^{-6}}} \end{aligned}$$

③ probability of (4 minutes or less)

$$\begin{aligned} p(t \leq 4) &= \int_0^4 3e^{-3t} dt = 3 \left[\frac{e^{-3t}}{-3} \right]_0^4 \\ &= -[e^{-3t}]_0^4 = -[e^{-12} - 1] \\ &= \underline{\underline{1 - e^{-12}}} \end{aligned}$$

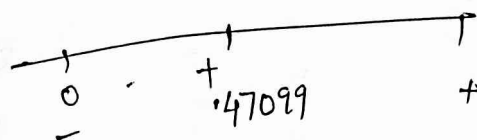
Module 4.

17 a)

Solve the equation $3x + \sin x - e^x = 0$ by
Regula Falsi Method.

Ans

$$f(x) = 3x + \sin x - e^x$$



$$f(0) = -1$$

$$f(1) = 3 + \sin 1 - e = \cancel{0.299} > 0$$

$\therefore$ Root lies b/w 0 and 1, $\therefore a=0$ $b=1$

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$= \frac{0 \times f(1) - 1 f(0)}{f(1) - f(0)} = \frac{-1}{0.299 - 1}$$

$$= 0.47099$$

$$f(x_1) = > 0$$

$\therefore$ Root lies b/w 0 and 0.47099.

$$a=0 \quad b=0.47099.$$

$$x_2 = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$

$$= 0.372277.$$

$$f(x_2) = f(0.372277) > 0.$$

$\therefore$ Root lies b/w 0 and 0.372277.

$$a=0 \quad b=0.372277$$

$$x_3 = \frac{af(b) - bf(a)}{f(b) - f(a)} = 0.361598$$

$$f(x_3) > 0$$

$\therefore$ Root lies b/w 0 and 0.361598.

$$x_4 = 0.360537.$$

$$x_5 = 0.360433.$$

$$x_6 = 0.360423$$

$$x_7 = 0.360422$$

$$x_8 = \underline{\underline{0.360422}}$$

17b)

Using Lagrange's interpolation method find $y(4)$, Given the following data.

x	1	3	5	7
y	24	120	336	720

Ans)

Lagrange's interpolation formula is

$$y = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n$$

$$x_0 = 1, \quad x_1 = 3 \quad x_2 = 5 \quad x_3 = 7. \quad x = 4.$$

$$y_0 = 24 \quad y_1 = 120 \quad y_2 = 336 \quad y_3 = 720.$$

$$y = \frac{(4-3)(4-5)(4-7)}{(1-3)(1-5)(1-7)} 24 + \frac{(4-1)(4-5)(4-7)}{(3-1)(3-5)(3-7)} 120 +$$

$$\frac{(4-1)(4-3)(4-7)}{(5-1)(5-3)(5-7)} 336 + \frac{(4-1)(4-3)(4-5)}{(7-1)(7-3)(7-5)} 720.$$

$$= \underline{\underline{210}}$$

18 a) Evaluate $y(2)$ from the following data using Newton's forward difference method.

x	0	5	10	15
y	14	379	1444	3584

ans) $p = \frac{x - x_0}{h}, \frac{x - 0}{5} = \frac{x}{5}, x = 2, p = \frac{2}{5}$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	14			
5	379	365	700	375
10	1444	1065	1075	
15	3584	2140		

$$y = y_0 + \frac{p}{1!} \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$$

=

$$y = 100.$$

b) Find $y(3)$ using Newton's divided difference formula, Given

$$y(1) = -26, \quad y(2) = 12, \quad y(4) = 256.$$

$$y(6) = 844$$

Ans)

$$y(1) = -26 \Rightarrow x_0 = 1$$

$$y(2) = 12 \Rightarrow x_1 = 2$$

$$y(4) = 256 \Rightarrow x_2 = 4$$

$$y(6) = 844 \Rightarrow x_3 = 6$$

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
1	$\textcircled{-26}$	$\frac{12 - (-26)}{2 - 1} = \textcircled{38}$	$\frac{132 - 38}{4 - 1} = \textcircled{28}$	$\frac{43 - 28}{6 - 1} = \textcircled{3}$
2	12			
4	256	$\frac{256 - 12}{4 - 2} = 122$	$\frac{294 - 122}{6 - 2} = 43$	
6	844	$\frac{844 - 256}{6 - 4} = 294$		

$$y = f(x_0) + (x - x_0)\Delta f(x_0) + (x - x_0)(x - x_1)\Delta^2 f(x_0) + (x - x_0)(x - x_1)(x - x_2)\Delta^3 f(x_0).$$

To find $f_y(3)$, substitute $x=3$ in the above equation.

$$y(3) = -26 + (3-1)38 + (3-1)(3-2)28 + (3-1)(3-2)(3-4)3.$$

$$= -26 + 2 \times 38 + 2 \times 28 + 2(-1)3$$

$$= -26 + 76 + 56 - 6.$$

$$= 132 - 32.$$

$$= 100.$$

Module 5

19 a). Solve the equations $4x + 2y + z = 14$, $x + 5y - z = 10$, $x + y + 8z = 20$, using Gauss Seidel Method

Ans)

$$x = \frac{1}{4} [14 - 2y - z]$$

$$y = \frac{1}{5} [10 - x + z]$$

$$z = \frac{1}{8} [20 - x - y]$$

1st iteration

$$x^{(1)} = 3.5 \quad y^{(1)} = 1.3 \quad z^{(1)} = 1.9$$

2nd iteration

$$x^{(2)} = 2.375 \quad y^{(2)} = 1.905 \quad z^{(2)} = 1.965$$

3rd iteration

$$x^{(3)} = 2.056 \quad y^{(3)} = 1.982 \quad z^{(3)} = 1.995$$

4th iteration

$$x^{(4)} = 2.010 \quad y^{(4)} = 1.997 \quad z^{(4)} = 1.999$$

5th iteration

$$x^{(5)} = 2.002 \quad y^{(5)} = 1.999 \quad z^{(5)} = 2.0$$

6th iteration

$$x^{(6)} = 2.001 \quad y^{(6)} = 2.0 \quad z^{(6)} = 2.0$$

7th iteration

$$x^{(7)} = 2.0 \quad y^{(7)} = 2.0 \quad z^{(7)} = 2.0$$

19b) Using Euler's method, solve $\frac{dy}{dx} = x + y + xy$, given $y(0) = 1$, evaluate y at $x = 0.1$ by taking $h = 0.05$.

Ans)
$$\frac{dy}{dx} = x + y + xy = f(x, y).$$

$x_0 = 0, y_0 = 1$ (given $y(0) = 1$)
 $y_1 = y_0 + h f(x_0, y_0).$

$y(0.05) = y_1 = 1 + 0.05 f(0, 1).$

$y(0.05) = 1.05$.

$y_2 = y_1 + h f(x_1, y_1) = 1.05 + 0.05 f(0.05, 1.05)$
 $= \underline{\underline{1.05275}}.$

209) Apply 4th order Runge Kutta Method to find $y(0.1)$ for the initial value problem

$\frac{dy}{dx} = x + y^2, y(0) = 1$

$$y' = \frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

$$y_{n+1} = y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = h f(x_n, y_n).$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right).$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + k_2\right)$$

$$k_4 = h f(x_n + h, y_n + k_3).$$

$$x_0 = 0 \quad y_0 = 1 \quad h = 0.1.$$

$$k_1 = 0.1.$$

$$\frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) = 0.11649.$$

$$k_2 = 0.11585$$

$$k_3 = 0.116857$$

$$k_4 = 0.1347.$$

$$y(0.1) = y_0 + 0.11649 = 1 + 0.11649 = \underline{\underline{1.1165}}.$$

20 b). Solve $\frac{dy}{dx} = x^2(1+y)$ for $x = 1.4$ using Adam's-Moulton Method, given $y(1) = 1$, $y(1.1) = 1.2333$

$$y(1.2) = 1.548 \quad y(1.3) = 1.979.$$

$$f_0 = f(x_0, y_0) = f(1.1) = 2.$$

$$f_1 = f(x_1, y_1) = f(1.1, 1.233) = 2.70193.$$

$$f_2 = f(x_2, y_2) = f(1.2, 1.548) = 3.66912.$$

$$f_3 = f(x_3, y_3) = f(1.3, 1.979) = 5.03451$$

$$y_4^* = 2.5723.$$

$$\text{corrector } f_4^* = f(1.4, 2.5723) = 7.0017.$$

$$y_4 = y(1.4) = \underline{\underline{2.574947}}.$$